SNAPGraph NC: Modeling the Tradeoff Between Payment Accuracy and Access in North Carolina SNAP Recertification

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Abstract

The Supplemental Nutrition Assistance Program (SNAP) requires households to recertify their eligibility at the end of each certification period, usually every 6 to 12 months, in order to continue receiving benefits. Stronger verification can help reduce payment errors, but additional requirements may also create barriers for eligible households. This tradeoff is especially important in North Carolina, where the state's SNAP payment error rate is above the 6% threshold at which states will begin paying a share of benefit costs under H.R. 1. Although previous research has examined SNAP participation and administrative burden, less attention has been given to how risk is distributed across individual steps in the recertification process. This project explored how checkpoint complexity, documentation burden, and the consequences of errors affect the balance between payment accuracy and household access. Rather than using actual case records, SNAPGraph NC was developed as a source-supported mathematical model. Ten recertification checkpoints were scored using a common rubric and represented as nodes in a directed graph. Ten hypothetical household paths were then used to calculate checkpoint risk, path risk, and risk concentration. A payoff matrix compared four administrative strategies, while a sensitivity analysis tested whether the main rankings changed under different weighting systems. Income verification had the highest total risk and concentration score, while the path involving variable income and work-rule screening had the highest cumulative risk. Under the assumed strategy multipliers, access-supportive targeted review produced the lowest average modeled risk. Income verification, document verification, and case-worker review also remained the highest-ranked concentration checkpoints across all three weighting versions. Overall, the model suggests that targeted review combined with household support at the most heavily used checkpoints may provide a better balance between reducing payment errors and protecting access than broad verification.

Introduction

A household can remain fully eligible for food assistance and still lose benefits because of a late form, a missing document, or a misunderstood rule. In the Supplemental Nutrition Assistance Program (SNAP), these losses happen most often at recertification, the periodic review in which households show that they still qualify and that their benefit amount is correct (Gray, 2019). Recertification protects program integrity, but each step also adds paperwork, deadlines, and verification that households must complete on time (Homonoff & Somerville, 2021; Moynihan et al., 2015). Table 1 summarizes how the main recertification features serve both goals at once.

Recertification Feature	**Intended Purpose**	**Possible Access Risk**
Notices and deadlines	Prompt households to complete recertification on time	Late, missed, or misunderstood notices may interrupt benefits
Income and document verification	Confirm eligibility and calculate the correct benefit amount	Missing or unclear records may delay or end benefits
Rule and exemption screening	Apply work requirements and determine whether exemptions apply	Complex rules may increase confusion or incorrect decisions
Benefit decision and notice	Communicate whether benefits continue or change	Delayed or unclear decisions may make corrections or appeals harder

Table 1: Examples of How SNAP Recertification Features Can Affect Payment Accuracy and Household Access

In North Carolina, this issue has become more pressing because the state's SNAP payment error rate remains above the 6% federal threshold that will trigger state cost-sharing beginning in October 2027 (NCDHHS, n.d.-a; USDA FNS, 2026). This creates pressure to tighten verification, which could reduce errors while making it harder for eligible households to stay enrolled. This

study asks how structural features of North Carolina's SNAP recertification process shape the balance between payment accuracy and access, and at which steps that balance is most at stake.

Background

SNAP and Its Administration

SNAP is the largest federal food assistance program in the United States. It provides monthly benefits on an Electronic Benefit Transfer (EBT) card that households use to buy groceries, with the goal of reducing food insecurity among low-income households. Eligibility depends mainly on household size and income, with additional rules for certain groups, such as able-bodied adults without dependents (Certification of Eligible Households, 2026).

The program is federally funded but jointly run. The USDA Food and Nutrition Service sets national rules and has paid the full cost of benefits, while states share administrative costs and handle applications and casework. In North Carolina, SNAP is called Food and Nutrition Services. NCDHHS supervises the program, and county Departments of Social Services (DSS) determine eligibility and manage cases. More than 1.4 million North Carolinians receive benefits each month (NCDHHS, n.d.-a, n.d.-c).

The Recertification Process

Households are approved for a fixed certification period, generally up to 12 months, or up to 24 months when all adult members are elderly or disabled. Before the period ends, the state must send a notice explaining how to renew. The household then submits a renewal application, completes an interview at least once every 12 months, and provides verification of information that has changed or is questionable, most often income (Certification of Eligible Households, 2026).

In North Carolina, the county DSS enters the recertification in NC FAST, the state's case management system, and schedules an interview if one has not been held in the past 12 months. If information is missing, the caseworker sends a Notice of Information Needed. For households that renew on time, benefits must be approved by the last workday of the certification period (NCDHHS, 2023).

If a household does not return the form, misses the interview, or does not provide requested documents in time, the case closes at the end of the period, even if the household is still eligible. Gray (2019) found that most new SNAP cases in Michigan left the program within a year, with exits clustered in the months when paperwork was due, and that many of those leaving still appeared eligible. Recertification is therefore the point where payment accuracy and access come into the most direct contact.

Payment Errors and Federal Oversight

Payment accuracy is measured through the SNAP Quality Control system. States review a random sample of active cases to check whether each household received the correct benefit amount, and federal reviewers re-check a subsample of those cases (Performance Reporting System, 2026). A payment error is an overpayment or underpayment above a small tolerance level. These errors are generally unintentional and come from case processing mistakes or household reporting issues, not fraud (NCDHHS, n.d.-a).

For fiscal year 2025, North Carolina had a 7.36% SNAP payment error rate, compared to the national average of 10.62% (USDA FNS, 2026). Despite being ahead of the national average, the state still exceeded 6%, the threshold that matters under H.R. 1, the federal budget law signed in July 2025. Beginning in October 2027, states with error rates of 6% or higher must pay a share of

SNAP benefit costs, up to 15%. For North Carolina, NCDHHS estimates that exposure at as much as $420 million per year (NCDHHS, n.d.-a).

H.R. 1 also lowers the federal share of administrative costs from 50% to 25% starting in October 2026 (NCDHHS, n.d.-a). In addition, it expands work requirements to many SNAP recipients aged 18 to 64 unless they qualify for an exemption, which adds a screening step to recertification (NCDHHS, n.d.-b).

Why Administrative Burden Persists

Moynihan et al. (2015) define administrative burden as the learning, psychological, and compliance costs people face when dealing with government. Research shows that these costs reduce participation in SNAP. Homonoff and Somerville (2021) found that households assigned later recertification interviews, which left less time to reschedule a missed interview, were 22% less likely to recertify successfully, even though the households who dropped off were as needy as the average participant. Kabbani and Wilde (2003) found that when states shortened certification periods to lower error rates, fewer eligible households participated, and Potluri et al. (2025) found that state policies that make participation easier are associated with lower food insecurity.

That burden reduces participation is therefore well established. The more useful question is why burdensome procedures persist when their costs are known. Herd and Moynihan (2018) argue that burden is often a policy choice rather than an accident, shaped by beliefs about who deserves help and by concerns about fraud and misuse of public funds.

In SNAP, the incentive structure points the same way. Quality Control, and now H.R. 1, attach financial penalties to payment errors, while incorrect denials and case closures are measured separately and carry no comparable penalty. A state under pressure to lower its error rate therefore has a reason to add verification even when that verification causes eligible households to lose

benefits. Reducing burden without ignoring these concerns requires knowing where in the process verification does the most to prevent errors and where it does the most to block eligible households.

Most previous studies analyze one aspect of administrative burden or compare different states' administrative policies. This paper instead uses the SNAPGraph NC network to model recertification as a connected series of steps and checkpoints where households can be delayed, redirected, or dropped. The goal is to identify the points where both kinds of risk are most concentrated.

Methodology

SNAPGraph NC is a source-supported mathematical model, not an empirical analysis of actual SNAP cases. Its scores are informed by federal and state program rules and prior research, but they are my own assumptions, not values measured by USDA or NCDHHS. I built the model in Excel and analyzed it in Google Colab, a free browser-based Python environment, using four Python libraries: pandas and NumPy for organizing data and performing calculations, NetworkX for building and analyzing the graph, and matplotlib for creating figures.

Graph theory is the branch of mathematics that studies networks made of nodes (points) connected by edges (links). In a directed graph, each edge has a direction, so it can represent a move from one stage to the next, and a path is an ordered sequence of nodes connected by edges. This structure fits recertification because the process is sequential, different households pass through different combinations of steps, and a problem at one step affects every step after it. Graph measures can show which steps the most paths depend on, which a simple list of steps cannot.

In SNAPGraph NC, each verification or decision step is a node, each move from one step to the next is an edge, and each household's route through the process is a path. I call these steps checkpoints. This is my own term, not an official SNAP term, chosen because each step is a point

where a case can either move forward or stall. I derived the ten checkpoints from the recertification requirements described above, grouping related tasks so that each node represents one distinct decision: recertification notice (C1), household composition (C2), income verification (C3), deduction and expense review (C4), work-requirement screening (C5), document verification (C6), case-worker or system review (C7), final benefit calculation (C8), notice of decision (C9), and outcome (C10).

Each checkpoint received three scores from 1 to 5: complexity, documentation burden, and consequence (Table 2). I assigned the scores using a common rubric, described in the note below Table 2, based on what federal regulations and the NCDHHS manual require at each step and on the burdens described in prior research. The scores are ordinal judgments, so a one-point difference indicates a higher or lower level rather than a measured quantity.

Checkpoint Code	**Checkpoint**	**Complexity**	**Documentation Burden**	**Consequence**
C1	Recertification notice	3	2	5
C2	Household composition	4	3	4
C3	Income verification	5	5	5
C4	Deduction and expense review	4	4	4
C5	Work-requirement screening	5	4	5
C6	Document verification	4	5	5
C7	Case-worker or system review	4	2	5

C8	Final benefit calculation	5	1	5
C9	Notice of decision	3	2	5
C10	Outcome	2	1	5

Table 2: SNAP Recertification Checkpoints and Modeled Rubric Scores

Note. Each dimension is scored from 1 (low) to 5 (high). Complexity reflects how many rules or calculations apply at that step. Documentation burden reflects the effort needed to obtain and submit what the step requires, not the number of pages: 1 = no household action, 2 = reading or responding to a notice, 3 = one or two documents the household usually has on hand, 4 = several documents or a document from a third party, and 5 = multiple third-party documents that may require contacting employers or landlords, travel, or waiting. Consequence reflects how serious an error at that step would be for benefit accuracy or continued access.

In this model, risk means the likelihood that a checkpoint either produces a payment error or burdens a household enough to interrupt access to benefits. Each checkpoint receives two component scores calculated from its Complexity (C), Documentation Burden (D), and Consequence (K) values:

Payment-error risk =

Access-burden risk =

Total checkpoint risk = Payment-error risk + Access-burden risk

Complexity was weighted most heavily for payment-error risk because complex rules and calculations create more opportunities for mistakes by caseworkers and households. Documentation burden was weighted most heavily for access-burden risk because missing or late documents are a common reason eligible households lose benefits. Consequence received the same

weight in both formulas because a serious error matters for both accuracy and access. These weights are assumptions, so I later tested whether the results changed under alternative weights.

I then developed 10 hypothetical household scenarios, each scored from 0 to 3 on six severity factors: income change, household composition, deductions, work-rule screening, document risk, and deadline or notice risk (Table 3). The severity scores guided which checkpoints each scenario's path included. A path's total risk was then calculated as the sum of the total checkpoint risk of every checkpoint on that path.

Path ID	Scenario	Income Change	H.C	D/E	W.R	D.R	Deadline/ Notice
P1	Stable Renewal	0	0	0	0	1	1
P2	Changing income household	3	1	1	0	3	2
P3	Work-Rule Adult	2	0	0	3	3	2
P4	Parent with children	1	3	2	0	3	2
P5	Elderly/Disabled household	1	2	3	0	3	2
P6	Missing documents	2	1	1	1	3	3
P7	Variable income + work-rule adult	3	1	2	3	3	3
P8	Household composition change	1	3	1	0	2	2
P9	Deduction-heavy household	1	1	3	0	3	2
P10	Notice or deadline-risk household	1	0	0	0	2	3

Table 3: Severity Scores Used to Construct the Ten Modeled Household Paths

Note. H.C. = household composition, D/E = deduction/expense, W.R. = work-rule screening, D.R. = document risk. Scores range from 0 (not a major issue in this path) to 3 (a high issue central to that path)

Finally, I calculated each checkpoint's path frequency, the number of scenarios whose path included that checkpoint. I then calculated a custom concentration score using the following formula:

Concentration score = Total checkpoint risk × Path frequency

Each path was recorded as an ordered sequence of checkpoints. Figure 1 shows the resulting network, where node size represents total risk and edge thickness represents transition frequency.

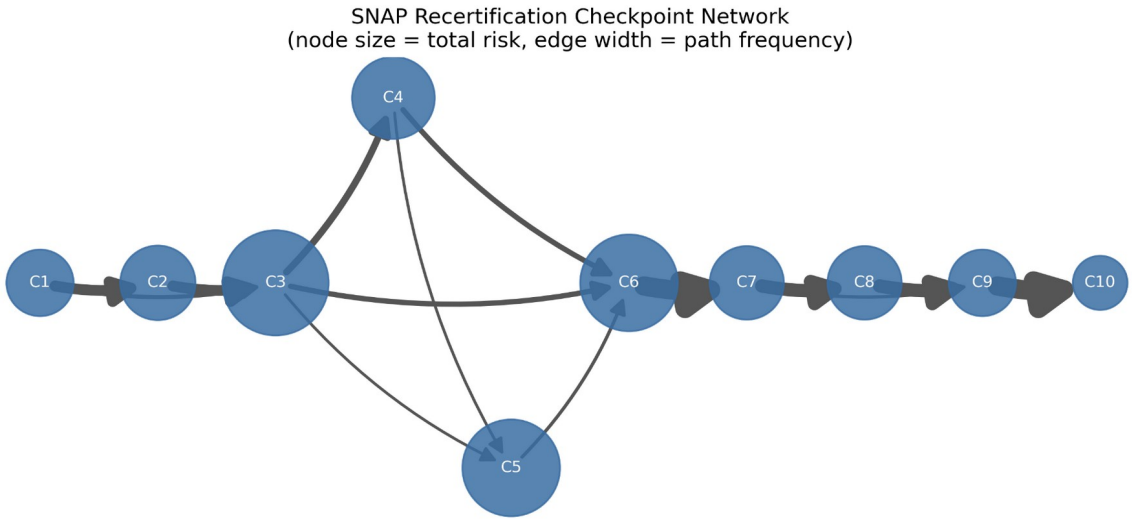


Figure 1: Directed Network of SNAP Recertification Checkpoints

The four administrative strategies were compared by applying different multipliers to payment-error risk and access-burden risk, as shown in Table 4. Broad enhanced verification adds

checks for every household, while targeted risk-based review focuses extra checks on higher-risk cases. The access-supportive strategy pairs targeted review with reminders, clearer notices, help with documentation, and follow-up. The multipliers are assumptions that represent the expected direction of each strategy's effect, not measured effects. A payoff was calculated for each strategy and household path, and the averages were used to compare strategies.

I also tested three weighting versions: the original weights, a complexity-heavy version, and a documentation-heavy version. Rankings that remained the same across all three versions were treated as more stable.

Strategy	**Modeled calculation**
Baseline recertification	1.00(payment-error risk) + 1.00(access-burden risk)
Broad enhanced verification	0.90(payment-error risk) + 1.25(access-burden risk)
Targeted risk-based review	0.85(payment-error risk) + 1.05(access-burden risk)
Access-supportive targeted review	0.90(payment-error risk) + 0.80(access-burden risk)

Table 4: Administrative Strategies Used in the Payoff Matrix

Results

All values reported come from the model's calculated output workbook. The model showed that checkpoint risk was concentrated in a small number of steps. Income verification (C3) had the highest total checkpoint risk at 10.0, followed by document verification (C6) and work-requirement screening (C5) at 9.3 each. Deduction and expense review (C4) followed at 8.0. Outcome (C10) had the lowest total risk at 5.1, while recertification notice (C1) and notice of decision (C9) each scored 6.5. Figure 2 presents the total-risk scores for all ten checkpoints.

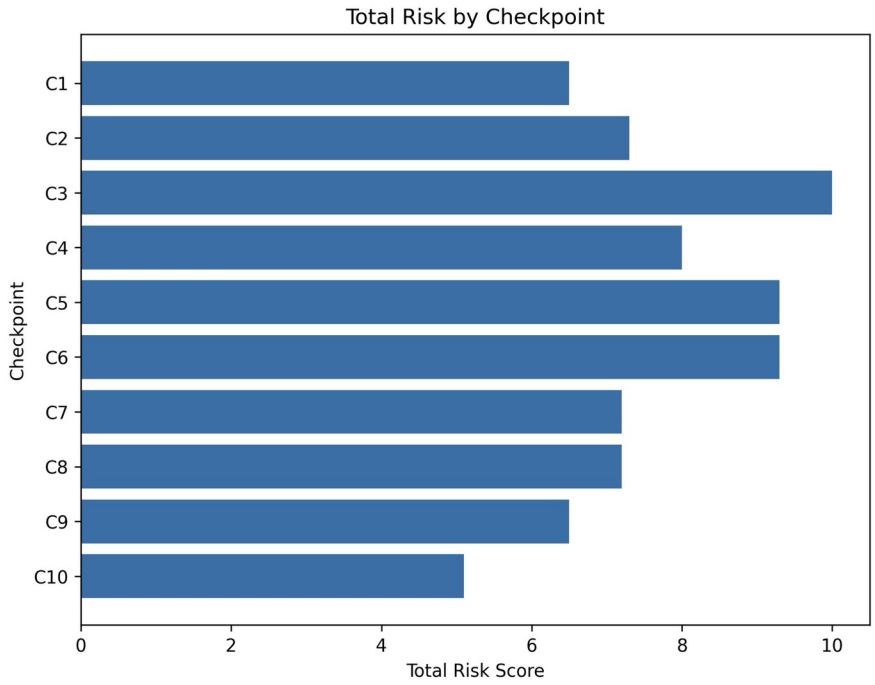


Figure 2: Total Modeled Risk by Checkpoint

Figure 3 displays the risk concentration by checkpoint. Weighting total risk by path frequency changed the rankings. C3 had the highest concentration score at 100, followed by C6 at 93 and C7 at 72. C5 fell to 18.6 because it appeared in only two paths, despite tying C6 for the second-highest individual risk.

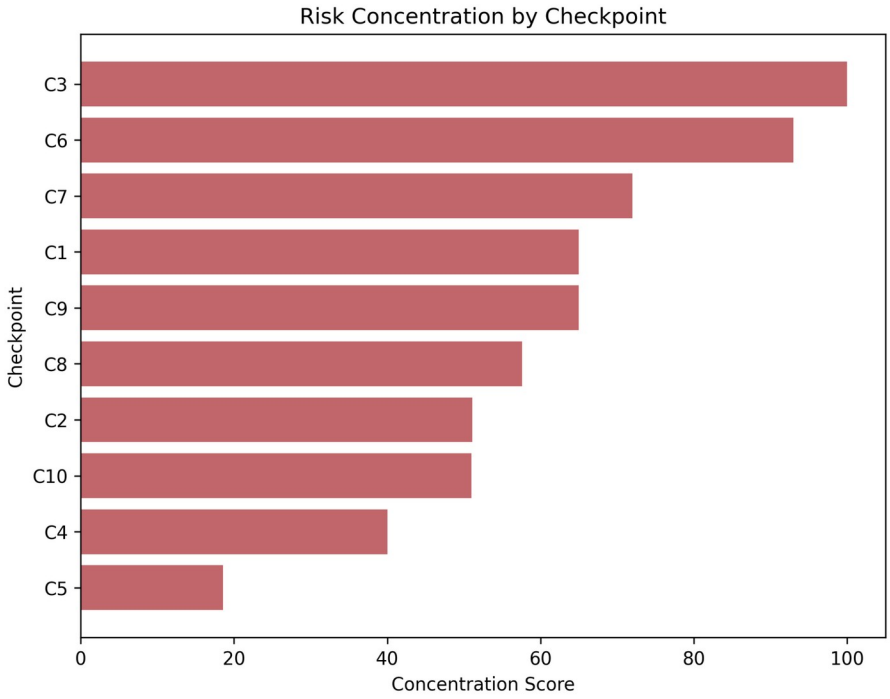


Figure 3: Risk Concentration by Checkpoint

Note. Concentration score equals total checkpoint risk multiplied by path frequency, or the number of modeled household paths containing that checkpoint.

Figure 4 shows the total path risk for each household scenario. P7, the variable-income plus work-rule adult path, had the highest total path risk at 76.4 and was the only path containing all ten checkpoints. Four paths (P2, P4, P5, P9) tied for second at 67.1, while P6 and P10 had the lowest path risks at 44.6.

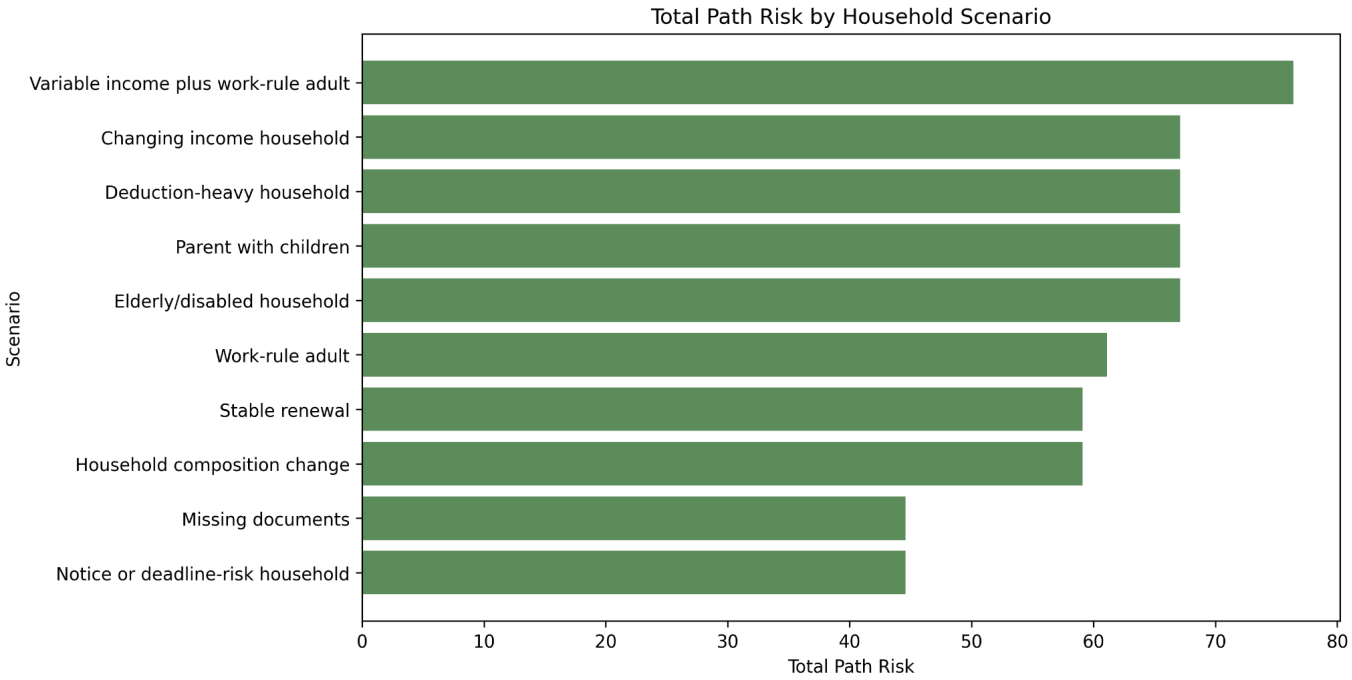


Figure 4: Total Path Risk by Household Scenario

Note. Each bar represents one hypothetical modeled path rather than an actual household.

The payoff matrix provided another use of the model by comparing administrative strategies. Across all ten household paths, access-supportive targeted review had the lowest average risk score at 52.25. Targeted risk-based review followed with an average score of 58.02, while baseline recertification had a higher average of 61.33. Broad enhanced verification had the highest average risk score at 65.50 and performed worse than the other strategies across every household path. Figure 5 shows the modeled risk scores for each administrative strategy and household path.

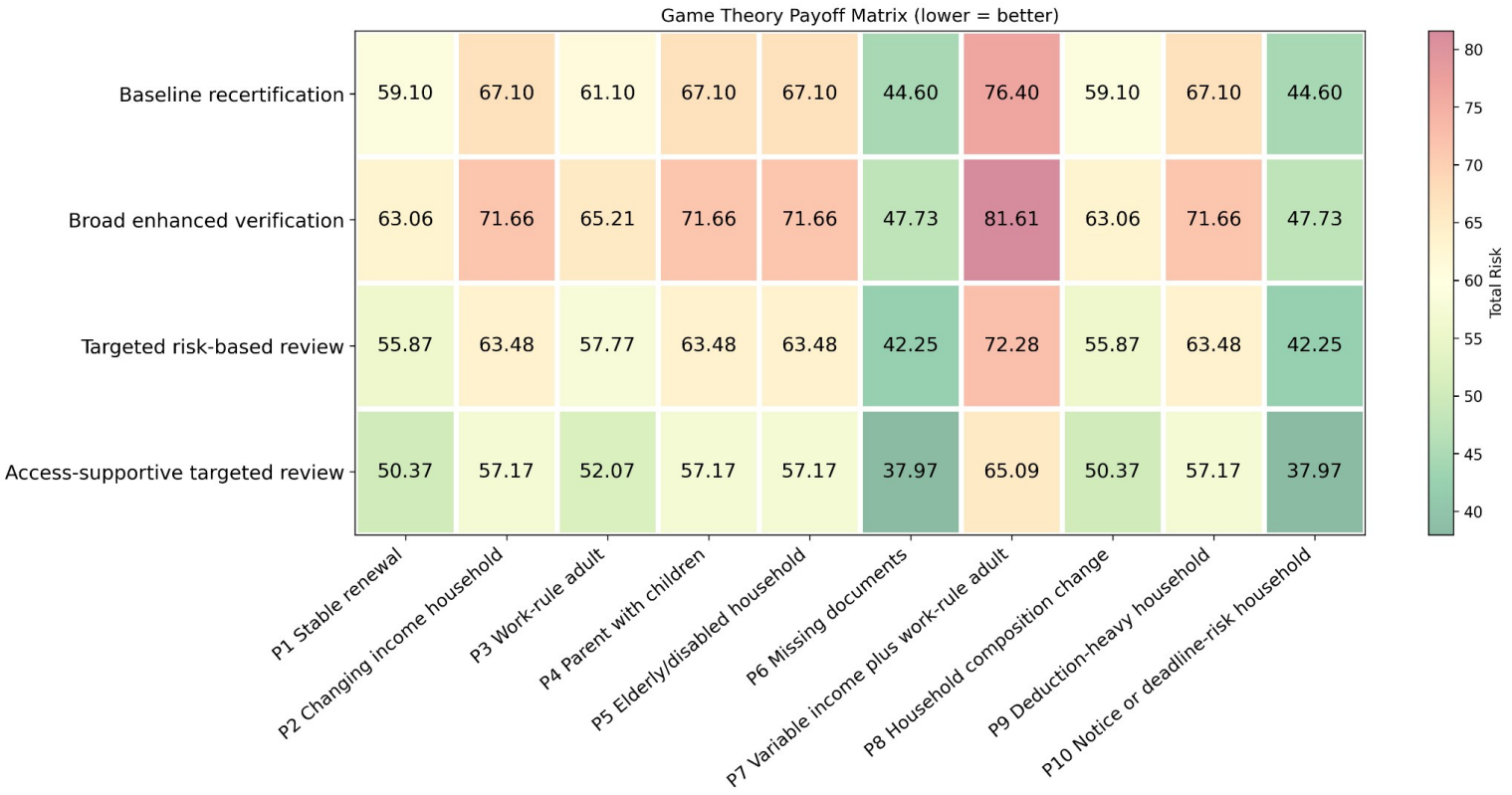


Figure 5: Administrative Strategy Payoff Matrix

Note. Lower values indicate better modeled performance. P represents a modeled household path.

Payment-error risk and access-burden risk varied by checkpoint. Final benefit calculation (C8) leaned toward payment-error risk (4.2 vs. 3.0), while document verification (C6) leaned slightly toward access-burden risk (4.5 vs. 4.8).

Finally, a sensitivity analysis was performed to test whether varying the weights would change the overall rankings. Figure 6 shows that changing the weights did not change the relative rankings, with C3 remaining the highest priority across all three versions, followed by C6 and then C7.

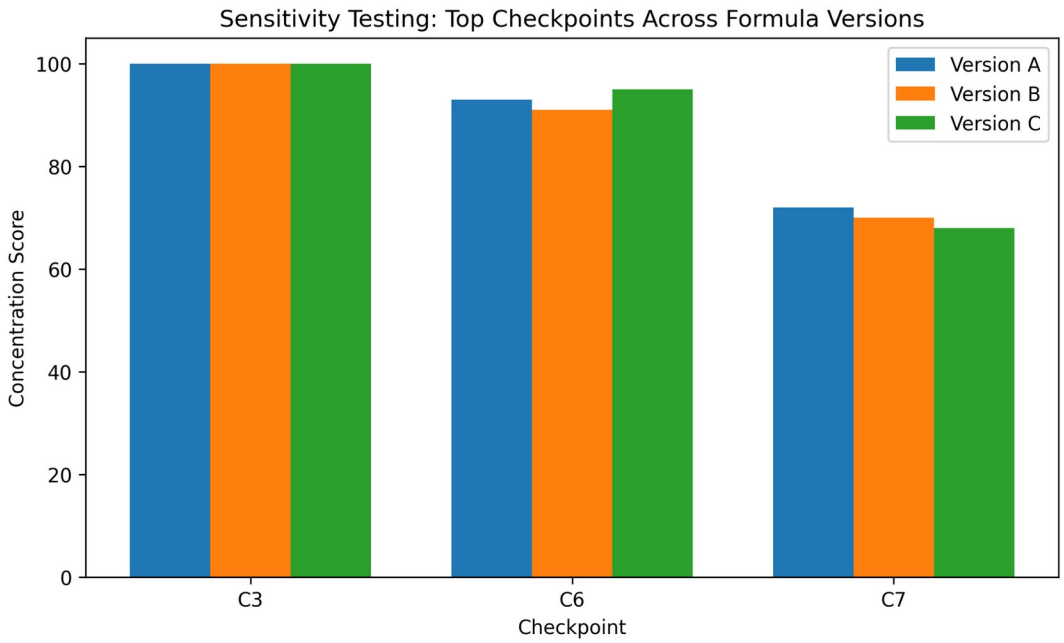


Figure 6: Sensitivity Testing Across Three Risk-Weighting Versions

Note. Version A uses the original weights, Version B gives complexity greater weight, and Version C gives documentation burden greater weight.

Discussion and Limitations

Within the model, recertification risk was concentrated at a few checkpoints that are both difficult and used by nearly every household. Income verification and document verification had the highest concentration scores because they combined high individual risk with presence in all ten paths. Work-requirement screening had a high individual score but a low concentration score because only two paths included it. This distinction matters for policy: a checkpoint that is risky for a small group calls for targeted help for that group, while a checkpoint that every household passes through is where process changes would reach the most people.

Payment-error risk and access burden were also distributed differently across checkpoints. Final benefit calculation leaned toward payment-error risk, while document verification leaned

slightly toward access burden, which suggests that the two goals do not conflict equally at every step. P7 had the highest path risk because it was the only path containing all ten checkpoints. Because path risk is a sum, longer paths will always score higher in this model, so path rankings mainly show which household situations involve the most steps. Similarly, C3, C6, and C7 appear in every path, which helps explain why their rankings stayed stable across the three weighting versions.

The strategy comparison should be read with the same care. Broad enhanced verification produced the highest modeled risk because its assumed increase in access burden outweighed its assumed decrease in payment-error risk, while access-supportive targeted review produced the lowest. Because each multiplier is applied the same way to every path, this ordering follows directly from the assumed multipliers and would hold for any household path. The ranking therefore restates the assumption that support reduces burden rather than providing independent evidence for it. What the model adds is the size of the gap for different households: the difference between broad verification and access-supportive review was largest for the longest path, P7 (16.5 points), and smallest for P6 and P10 (9.8 points).

These results are consistent with prior research. Homonoff and Somerville (2021) and Gray (2019) show that more demanding recertification procedures reduce the share of eligible households who stay enrolled, and Potluri et al. (2025) link participation-friendly policies to lower food insecurity. That more burden reduces access is not surprising on its own. The more important question is why burden persists, and the incentive structure described earlier offers one answer: Quality Control penalties and H.R. 1 reward states for reducing payment errors but not for keeping eligible households enrolled (Herd & Moynihan, 2018; Kabbani & Wilde, 2003).

The model's contribution is to show where those incentives collide. Income verification and document verification are where added verification is most likely to reduce payment errors, but they also carry the highest access burden for the most households. A targeted approach would keep verification at those checkpoints while adding support there, for example by using electronic data matches to verify income without requiring a household-supplied document, extending deadlines, or helping households obtain records. This could help North Carolina move below the 6% threshold without shifting the cost onto eligible families.

Several limitations should be noted. First, the rubric scores and formula weights are my assumptions, informed by program rules and prior research but not measured, and the ten household paths are simplified compared with real cases. Second, path frequency reflects only the scenarios built into the model, not how often each situation occurs in North Carolina. Third, checkpoint inclusion was assigned holistically per scenario rather than by a fixed rule, so paths with identical scores on a given factor can occasionally differ in which checkpoints they include.

Fourth, the documentation burden score cannot fully capture lived experience. A single request can be far harder than its score suggests: obtaining an official record, such as a certified copy of a birth certificate, or a statement from an employer who is slow to respond can take weeks of emails, travel, appointments, and fees (M. Berner, personal communication, August 9, 2026). The model also does not account for caseworker judgment, technology, communication problems, or processing delays, all of which shape what households experience.

Fifth, the strategy multipliers are assumptions, so the payoff matrix shows the consequences of those assumptions rather than measured effects. The sensitivity analysis confirms only that the checkpoint rankings were stable within this model, not that they would hold under

real-world conditions. For these reasons, the model cannot establish causal relationships or make accurate predictions about future payment error rates.

Future work should test the model against real experience before it is used to guide policy. This could include interviewing county DSS caseworkers and SNAP participants to check whether the rubric scores match what each checkpoint demands, weighting documentation burden by the time and cost of obtaining each document, applying the checkpoint structure to real administrative or case-level data, comparing North Carolina's process to other states, and testing the model against a larger and more detailed set of household paths.

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